



Data Communication & Network

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Network Classification



Network Classification

Classification by network geography

- According to the geographical boundaries spanned by the network itself
- This type of network classification is based on the **physical distance or area** covered by the network. It ranges from tiny personal networks to global-scale systems
- LAN, WAN, MAN ,PAN, SAN(Major two are LAN and WAN)

Classification by component roles/Network Architecture

- according to the roles that the networked computers play in the network's operation
- This classification focuses on how different devices (nodes) in a network interact and share responsibilities, especially in terms of resource management, control, and communication.
- Peer-to-peer, server-based, and client-based .



Classification by Network geography

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	
1 km	Campus	Local area network
10 km	City	
100 km	Country	Metropolitan area network
1000 km	Continent	
10,000 km	Planet	Wide area network
		The Internet



Classification by Network geography

1. PAN – Personal Area Network

Range: Within a few meters (typically ~10 meters)

Purpose: Connect personal devices for individual use

Technology: Bluetooth, Infrared, USB

Examples:

Connecting a smartphone to wireless earbuds

Smartwatch syncing with your phone

2. LAN – Local Area Network

Range: Within a building or campus (~100 meters)

Purpose: High-speed connection between computers and devices in a localized space

Technology: Ethernet, Wi-Fi

Examples:

Home or office Wi-Fi

School or university computer lab network



Classification by Network geography

3. CAN – Campus Area Network

Range: Multiple LANs connected within a campus or large site

Purpose: Link departments or buildings in a single organization

Technology: Ethernet, Fiber optics

Examples:

University campus with separate LANs for each faculty

Business park with interconnected office buildings

4. MAN – Metropolitan Area Network

Range: Covers a city or metro area (several kilometers)

Purpose: Connect LANs across a city for government, business, or ISP services

Technology: Fiber optics, leased lines

Examples:

City-wide cable TV or broadband networks

Municipal Wi-Fi projects



Classification by Network geography

5. WAN – Wide Area Network

Range: Covers a country, continent, or the entire globe

Purpose: Connects LANs and MANs over long distances

Technology: Satellites, leased telecom lines, undersea cables

Examples:

The Internet (largest WAN)

Bank networks spanning branches in multiple cities or countries

6. SAN – Storage Area Network

Range: Within a data center or enterprise environment (limited to building/campus scale)

Purpose: Provides high-speed, block-level access to centralized storage for servers

Technology: Fibre Channel

Examples:

Enterprise database servers accessing shared storage arrays

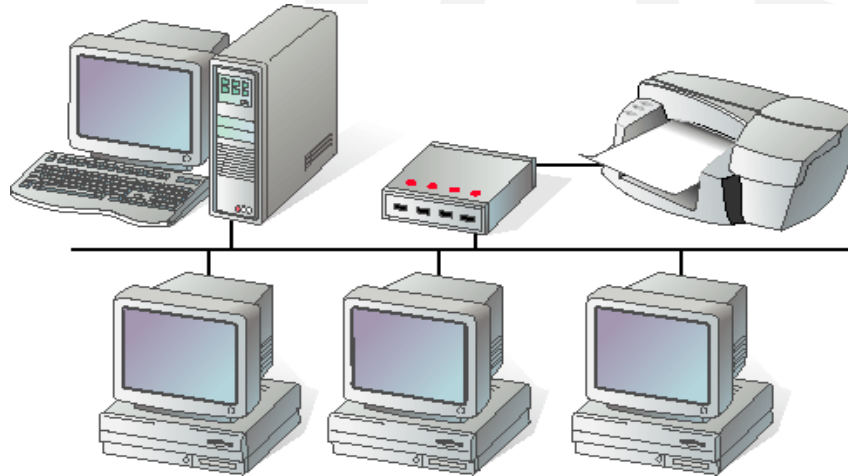
Virtualization platforms (e.g., VMware) using centralized SAN storage

Backup systems using SAN-connected tape libraries or disk arrays



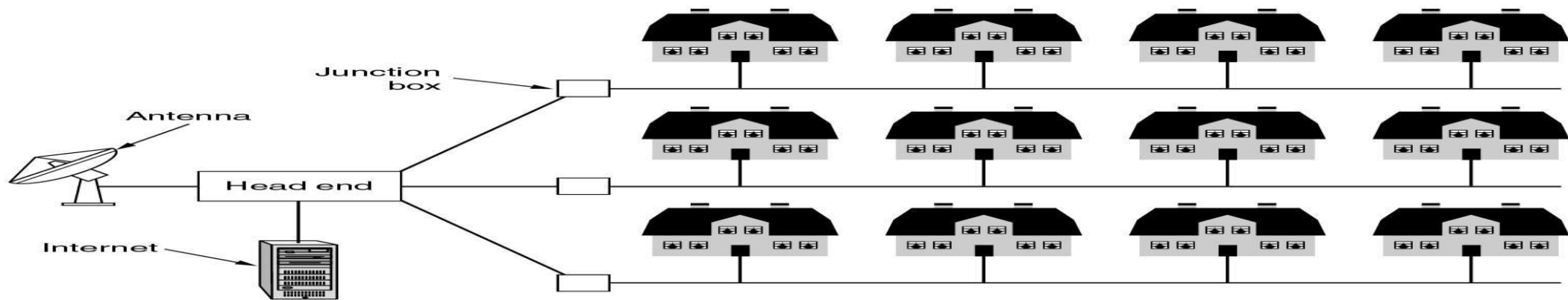
LAN (Local Area Network) : Wired Network

- Network in small geographical Area (Room, Building or a Campus)
- **Short distances (100 meters)**
- **Designed to provide local interconnectivity**
- LAN's can either be made wired or wireless. Twisted pair, coax or fiber optic cable can be used in wired LAN's
- a network that is used for communicating among computer devices, usually within an office building or home.



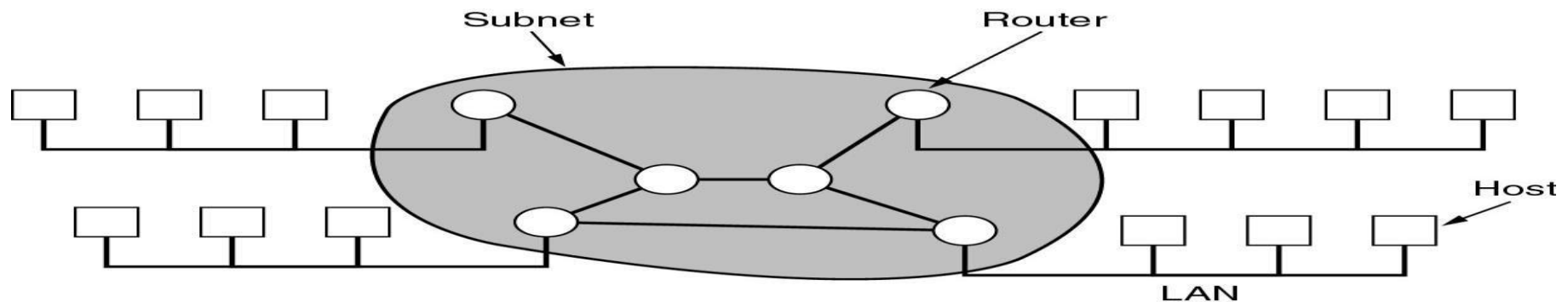
MAN

- A MAN spans the distance of a typical metropolitan city.
- The cost of installation and operation is higher.
- MANs use high-speed connections such as fiber optics to achieve higher speeds.
- Provide connectivity over areas such as a city, a campus
- More than 100m , Designed to handle data communication for multiple organizations in a city and nearby cities as well
- e.g. cable television network



WAN

- Network spread geographically (Country or across Globe)
- WANs consist of two distinct components:
 - transmission lines (copper, fiber, microwave) and switches (electronics, optics)
 - Store-and-forward or packet-switched subnet
- WANs span a larger area than a single city.
- These use long distance telecommunication networks for connection, thereby increasing the cost.
- The Internet is a good example of a WAN.
- More than 1000m long distance, Provide connectivity over large areas



Other Types of Network

CAN (Campus Area Network)

Eg. Sunbeam

PAN(Personal Area Network)

Laptop connected to Bluetooth, speaker, camera , mic etc made up of a wireless modem, a computer or two, phones, printers, tablets, etc., and revolves around one person in one building.

Private Area Network

E.g. One m/c in India (One IP) is connected to other machine in US (Other IP) are connected via Internet / VPN(Virtual Private Network)

SAN(Storage Area Network)

designed to provide high-speed connection in server-to-server applications (cluster environments)

Bank Branches and their transactions are done at centralized Bank server

That connects shared pools of storage devices to several servers, they don't rely on a LAN or WAN.

One instance of server is created and kept at each location

Google Drive data (google.co.in/ google.com/google.jp etc)

GAN(Global Area Network)

network composed of different interconnected networks that cover an unlimited geographical area.

used to support mobile number of arbitrary number of wireless LAN (LAN), satellite coverage areas, etc



Other Types of Network

- **VPN stands for Virtual Private Network.**
- It allows you to connect your computer to a private network, creating an encrypted connection that masks your IP address to securely share data and surf the web, protecting your identity online.
- A virtual private network, or VPN, is an encrypted connection over the Internet from a device to a network.
- The encrypted connection helps ensure that sensitive data is safely transmitted.
- It prevents unauthorized people and allows the user to conduct work remotely.
- VPN technology is widely used in corporate environments.

- **EPN (Enterprise Private Network)**
- Enterprise networking involves connecting various devices and components within an organization, such as data centers, branch offices, and cloud services, to facilitate data exchange, business processes, and analysis of network activity.
- Enterprise networking typically begins with a local area network (LAN), which connects devices within a particular location or facility. These LANs may then be connected to each other using a wide area network (WAN), which enables data to be transmitted over longer distances.
- The devices within an enterprise network are connected to a router or a switch, which serves as the central hub for network traffic.



BLUETOOTH

- Bluetooth is a wireless technology that lets devices like phones, tablets, and headphones connect to each other and share information without needing cables.
- Bluetooth follows the principle of transmitting and receiving data using radio waves.
- It is a Wireless Personal Area Network (WPAN) technology and is used for data communications over smaller distances.
- The spreading method that it uses is FHSS (Frequency-hopping Spread spectrum).
- A Bluetooth network is called a piconet (A small ad hoc network of up to 8 devices (1 master, 7 slaves)).
- A group of interconnected piconets is called a scatternet.

Q. A Bluetooth network consists of _____ primary device(s) and up to _____ secondary devices.

- A. one; five
- B. five; three
- C. two; six
- D. one; seven

Ans: D



Classification by Component Roles / Network Architecture

This classification focuses on **how different devices (nodes)** in a network **interact and share responsibilities**, especially in terms of resource management, control, and communication.

1. Peer-to-Peer (P2P) Architecture-

- All devices are equal (peers) and can act as both clients and servers.
- Peers share resources directly without a centralized server.
- Decentralized ,
- Less expensive (no dedicated server needed)
- Difficult to manage and secure in large scale
- Eg: File sharing ,LAN-based multiplayer gaming, Block chain and decentralized applications



Classification by Component Roles / Network Architecture

2. Client-Server Architecture -

- A central server provides resources and services.
- Clients (devices/users) request and consume those services.
- Centralized control and management
- Easier to secure and back up
- Scalable (can add more clients easily)
- Eg: Web browsing: your browser (client) connects to a web server, Email services, file servers, database access.



Ethernet Frame Format/MAC Frame

Preamble	SFD	Destination MAC	Source MAC	Type	Data and Pad	FCS
7 Bytes	1 Byte	6 Bytes	6 Bytes	2 Bytes	46-1500 Bytes	4 Bytes

Preamble

- informs the receiving system that a frame is starting and enables synchronization. In IEEE 802.3, eighth byte is start of frame (10101011)

SFD (Start Frame Delimiter)

- signifies that the Destination MAC Address field begins with the next byte.

Destination MAC

- identifies the receiving system.

Source MAC

- identifies the sending system.

Type

- defines the type of protocol inside the frame, for example IPv4 or IPv6.

Data and Pad

- contains the payload data.
- Padding data is added to meet the minimum length requirement for this field (46 bytes).

FCS (Frame Check Sequence)

- contains a 32-bit Cyclic Redundancy Check (CRC) which allows detection of corrupted data.



Network Physical Structure



Type of Connection

- Point to Point - single transmitter and receiver
- Multipoint - multiple recipients of single transmission

Physical Topology

- Connection of devices
- Refers to the way in which a network is laid out physically
- The geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.

Types of Connection

Point-to-Point Communication

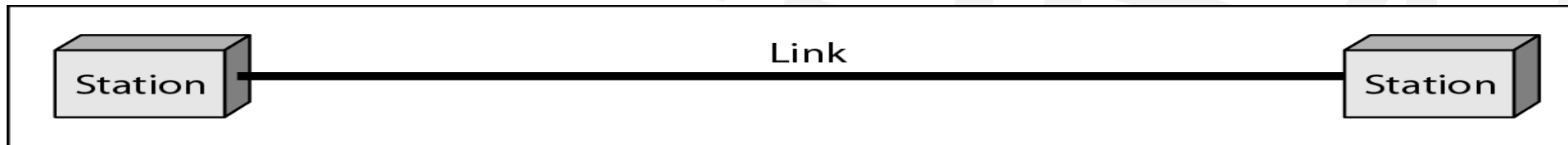
A direct connection between a single sender and a single receiver.

The communication path is dedicated between two devices.

High performance, low interference, and simpler management.

Eg: Computer connected to a modem via a cable, A leased line between two branch offices ,Bluetooth file transfer between two phones

Point-to-point: a dedicated link between two devices



a. Point-to-point



Types of Connection

Multipoint Communication

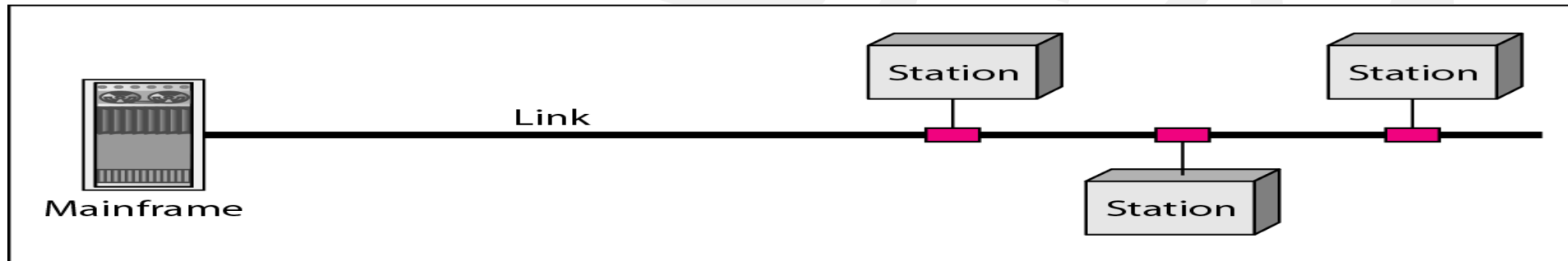
A single sender communicates with multiple receivers over the same medium.

The communication channel is shared among multiple devices.

Efficient use of resources in broadcast or shared environments.

Eg: LAN using a hub (all devices share the same medium), Cable TV network (single signal received by multiple homes) , Wireless broadcast from an access point to several devices

Multipoint: More than two specific devices share a single link



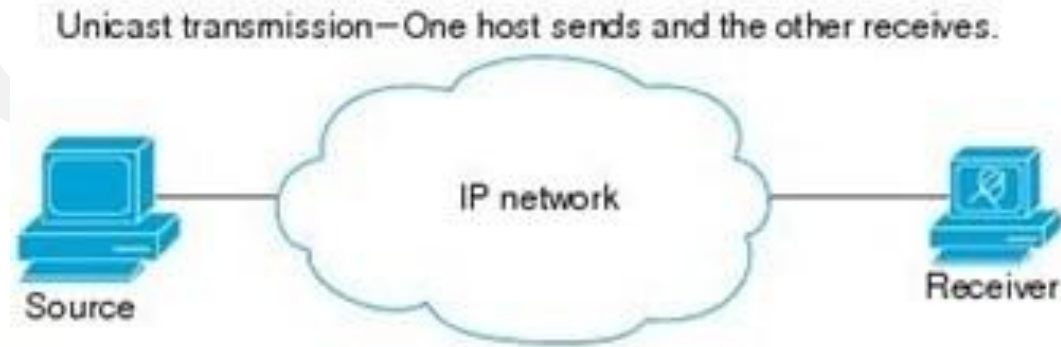
b. Multipoint



Type of transmission

Type of transmission - unicast, multicast, broadcast

1) Unicast Transmission

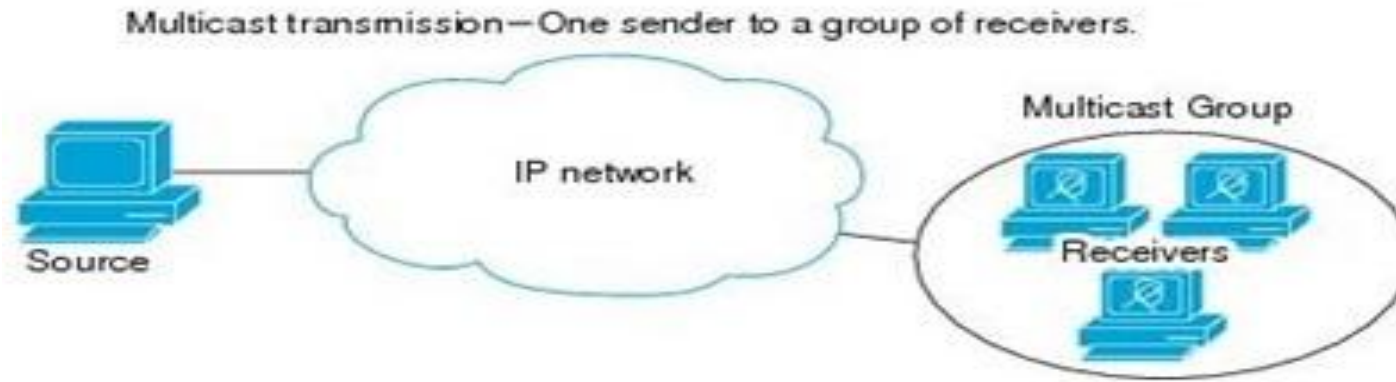


- The term unicast refers to "one to one".
 - In unicast mode, transmission happens from host to host i.e. one to one. This will not incur any traffic burden on the LAN.
 - The devices such as switch uses unicast mode of transmission.
- Eg: Phone call between two people



Type of transmission

2) Multicast Transmission

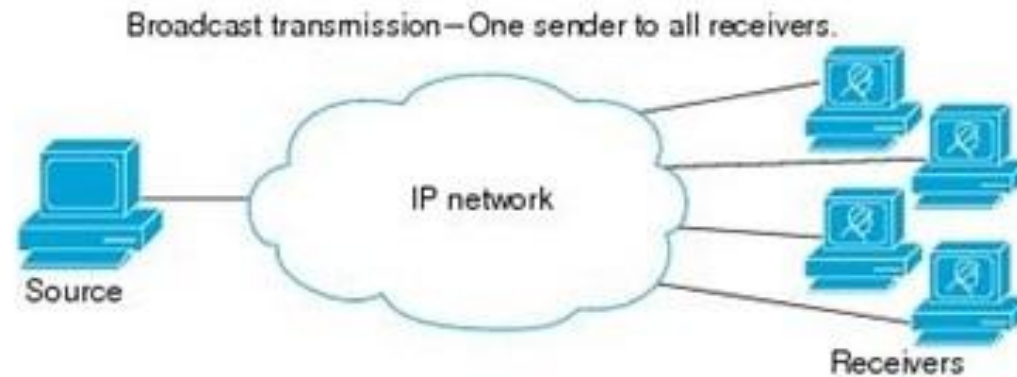


- The term **multicast** refers to "one to selected group of members".
 - In multicast mode, transmission happens from one host machine to selected group of host machines.
- Eg: giving a group call on WhatsApp or any application.



Type of transmission

3) Broadcast Transmission



The term **broadcast** refers to "one to all".

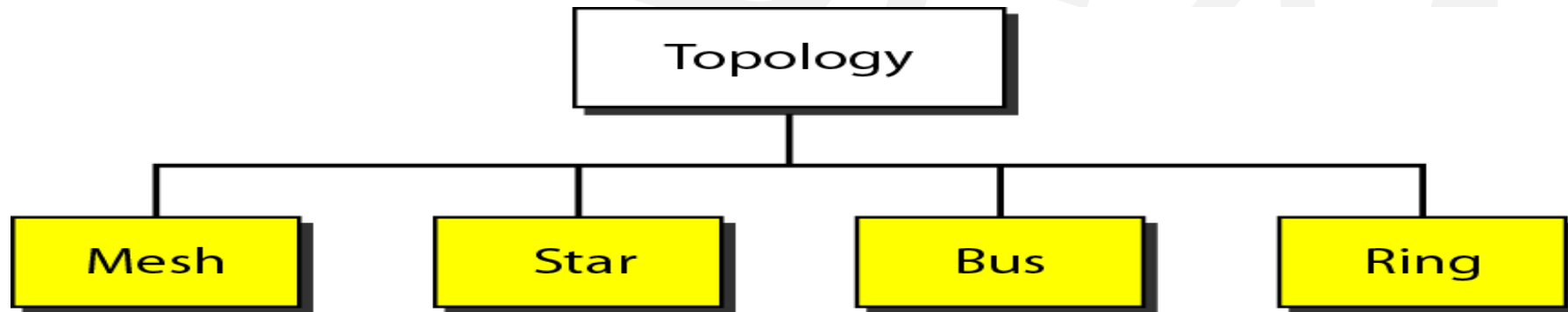
- In broadcast mode, transmission happens from one host to all the other hosts connected on the LAN.

The devices such as bridge uses this. The protocol such as ARP implement this, in order to know MAC address for the corresponding IP address of the host machine.

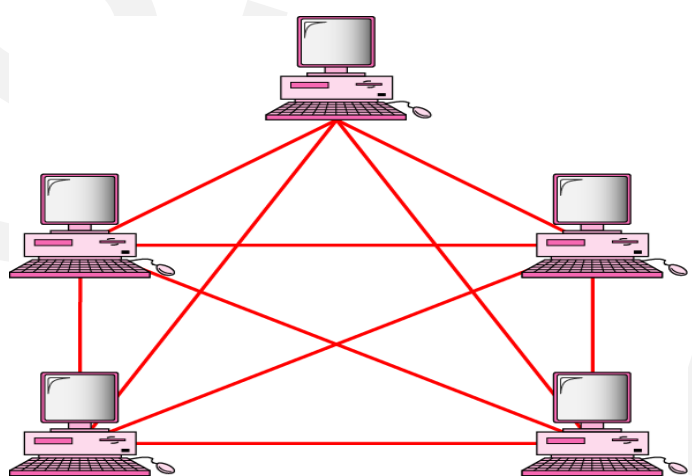


Physical Topology

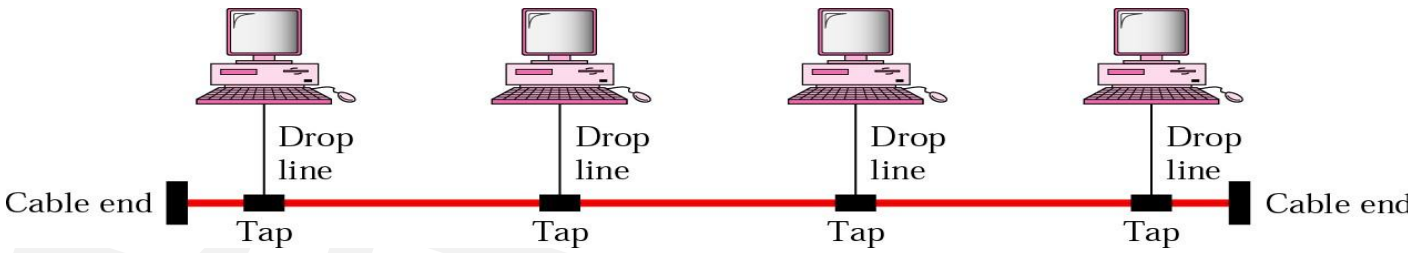
- Topology defines the way hosts are connected to the network
- The network topology defines the way in which computers, printers, and other devices are connected.
- A network topology describes the layout of the wire and devices as well as the paths used by data transmissions.
- Physical topology is the geometric representation of all the nodes in a network.



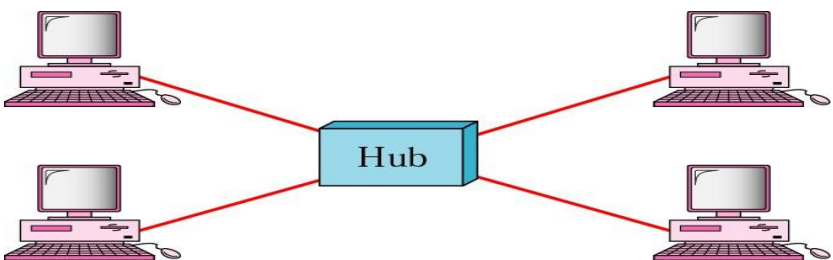
Network Topology



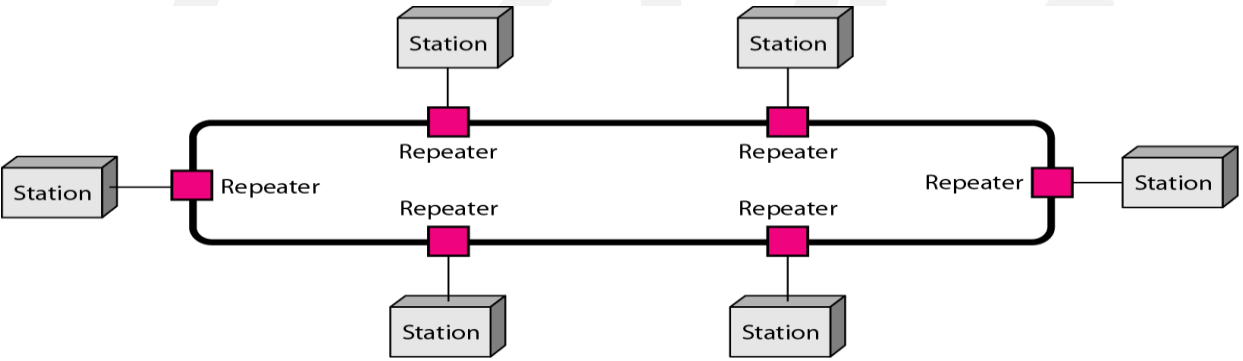
mesh



bus



star



ring



Mesh

- Mesh topology is a type of networking in which all the computers are inter-connected to each other.
- In Mesh Topology, the connections between devices take place randomly.
- The connected nodes can be computers, switches, hubs, or any other devices.
- In this topology, even if one of the connections goes down, it allows other nodes to be distributed.
- This type of topology is very costly.
- It is used for wireless networks, and its connections can be wired or wireless.
- There is a **point-to-point** connection between all nodes in the mesh topology setup.
- we have n devices in the network then each device must be connected with $(n-1)$ devices of the network.
- Number of links in a mesh topology of n devices would be $n(n-1)/2$.
- It is highly reliable and robust, often used in **critical communication systems**.
- Eg: internet backbone, various internet service providers are connected to each other via dedicated channels. Also used in military communication systems and aircraft navigation systems.



Star

- Star topology, sometimes known as a star network, is a network topology in which each device is connected to a central hub.
- In this network arrangement, all devices linked to a central network device are displayed as a star.
- In star topology, all connected devices are completely dependent on the central device; the communication through the whole Computer Network fails if the central device gets any problem.
- Each node in this diagram has a direct **point-to-point** link to the central device, yet no single node can communicate directly with the others. Therefore, before reaching the destination, each message has to pass through this central device only.
- Point-to-point connection between hosts and hub.
- Coaxial cables or RJ-45 cables are used to connect the computers.
- Eg: A local area network (LAN) in an office where all computers are connected to a central hub. Also used in wireless networks where all devices are connected to a wireless access point.
- Banking system every bank is connected to the RBI.



Bus

- The bus topology is designed in such a way that all the stations are connected through a single cable known as a **backbone cable**.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- The configuration of a bus topology is quite simpler as compared to other topologies.
- The backbone cable is considered as a "**single lane**" through which the message is broadcast to all the stations.
- In this topology, even if one of the connections goes down, it does not affect whole network but if backbone cable is affected then whole network is affected.
- There is a **multipoint connection** between all nodes.
- The most common access method of the bus topologies is **CSMA** (Carrier Sense Multiple Access).
- This method is used to prevent the collision of data as two or more devices can send the data to main cable at the same time.
- Eg: Ethernet LAN, where all devices are connected to a single coaxial cable or twisted pair cable. Also used in cable television networks.



Ring

- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.
- The data flows in one direction, i.e., it is **unidirectional**.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a **clockwise direction**.
- There is a **multipoint connection** between all nodes.
- But each system is connect in point-to-point fashion and if one system goes down then entire network will go down.
- Access method used in ring topology is **token passing method**. The word token describes segment of data send through the network. There are multiple tokens available on the network, the device that successfully acquires the token attaches the data to the token. The device that successfully decodes the token, receives the data.
- Eg: industrial control systems, telecommunications, and some security and transportation networks.



Network Devices / Internetworking Devices



Internetworking Devices

- Internetworking devices are products used to connect networks.
- As computer networks grow in size and complexity, so the internetworking devices used to connect them.
 - Hubs
 - Repeaters
 - Bridges
 - Switches
 - Routers
 - Gateways



Hubs

- Hub is used to build a LAN.
- Common connection point for devices in a network.
- It is non intelligent device.
- It does not understand the addressing.
- Hub is Multiport repeater containing multiple ports to interconnect multiple devices
- Hubs regenerate and retime network signals (increases traffic and collision)
- They cannot filter network traffic and they cannot determine best path
- The hub contains multiple ports.
- When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets.
 - does not concern about the address
 - concerns with only electrical signals
 - increases the traffic, as they broadcast data to all
 - increases the collision



Repeaters

- Repeaters or hubs work at the OSI **physical layer** to **regenerate the network's signal** and resend them to other segments.
- Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network.
- The longer the cable length, the weaker and more deteriorated the signals become as they pass along the networking media.
- Repeaters can be installed along the way to ensure that data packets reach destination.

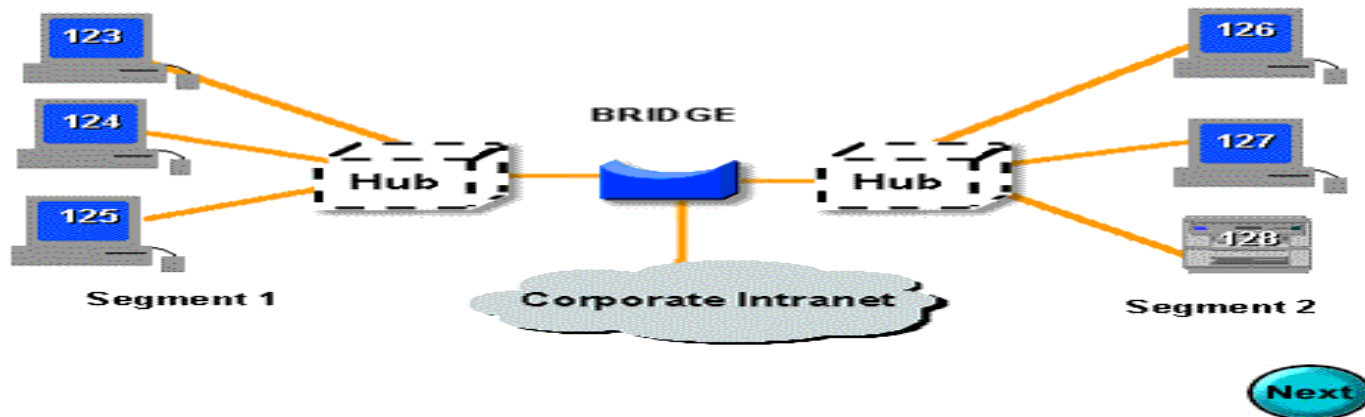
One way to solve the problems of too much traffic on a network and too many collisions is to use an internetworking device **called a bridge**.



Bridges : Operates at Data Link Layer

- A bridge eliminates unnecessary traffic and minimizes the chances of collisions occurring on a network by dividing it into segments .
- Device that connects and passes packets between two network segments.
- More intelligent than hub- As they analyze incoming packets and forwards (or drops) based on addressing information.(Routing Table is Build to record segment number of address)
- **Bridges work best where traffic from one segment of a network to other segments is not too great.**

Bridge Example



However, when traffic between network segments becomes too heavy, the bridge can become a bottleneck and actually slow down communication.



Switches (Multiport Bridges)

- **Switches operate at the Data Link layer (layer 2) of the OSI model**
- A switch is a device that is used to segment networks into sub networks called subnets. (Used to build LAN)
- **Can interpret address information**
- Uses Addressing Scheme known as MAC Addressing.
- Switches are capable of inspecting data packets as they are received, determining the source and destination device of that packet, and forwarding it appropriately
- Switch conserves network bandwidth and offers generally better performance than a hub.
- **Switch may Broadcast , unicast or Multicast .**

Learning the MAC Addresses and forwarding to the respective machine is switching.

- Switches have
 - ASIC (Application Specific IC)
 - OS is hardcoded in microprocessor
 - So switches are hardware based.
 - Ports are unlimited

- Bridges have
 - OS is separated
 - So bridges are not used
 - Bridges are software based.
 - Limited Ports (16)



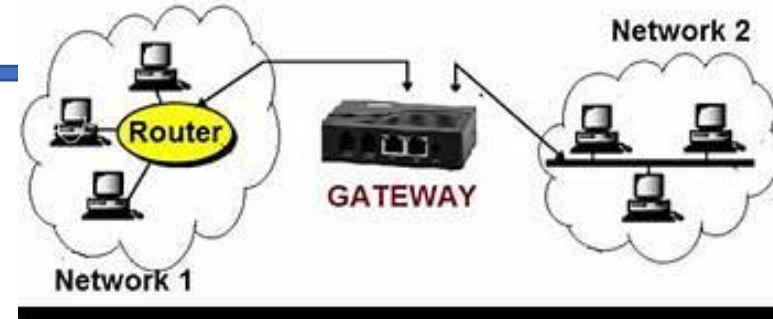
Routers

- Used to build WAN
- Router connect multiple networks and route the packets.
- Uses IP Address to identify every machine uniquely.
- Routers are used to connect two or more networks. For routing to be successful, each network must have a unique network number
- Routers have the ability to make intelligent decisions as to the best path for delivery of data on the network.
- **They use the “logical address” of packets and routing tables to determine the best path for data delivery.**
- To determine the **best path**, routers communicate with each other through **routing protocols**
- The four most common routing protocols:
 - RIP (Routing Information Protocol) for IP
 - OSPF (Open Shortest Path First) for IP
 - EIGRP (Enhanced Interior Gateway Routing Protocol) for IP, IPX, and AppleTalk
 - BGP (Border Gateway Protocol) for IP



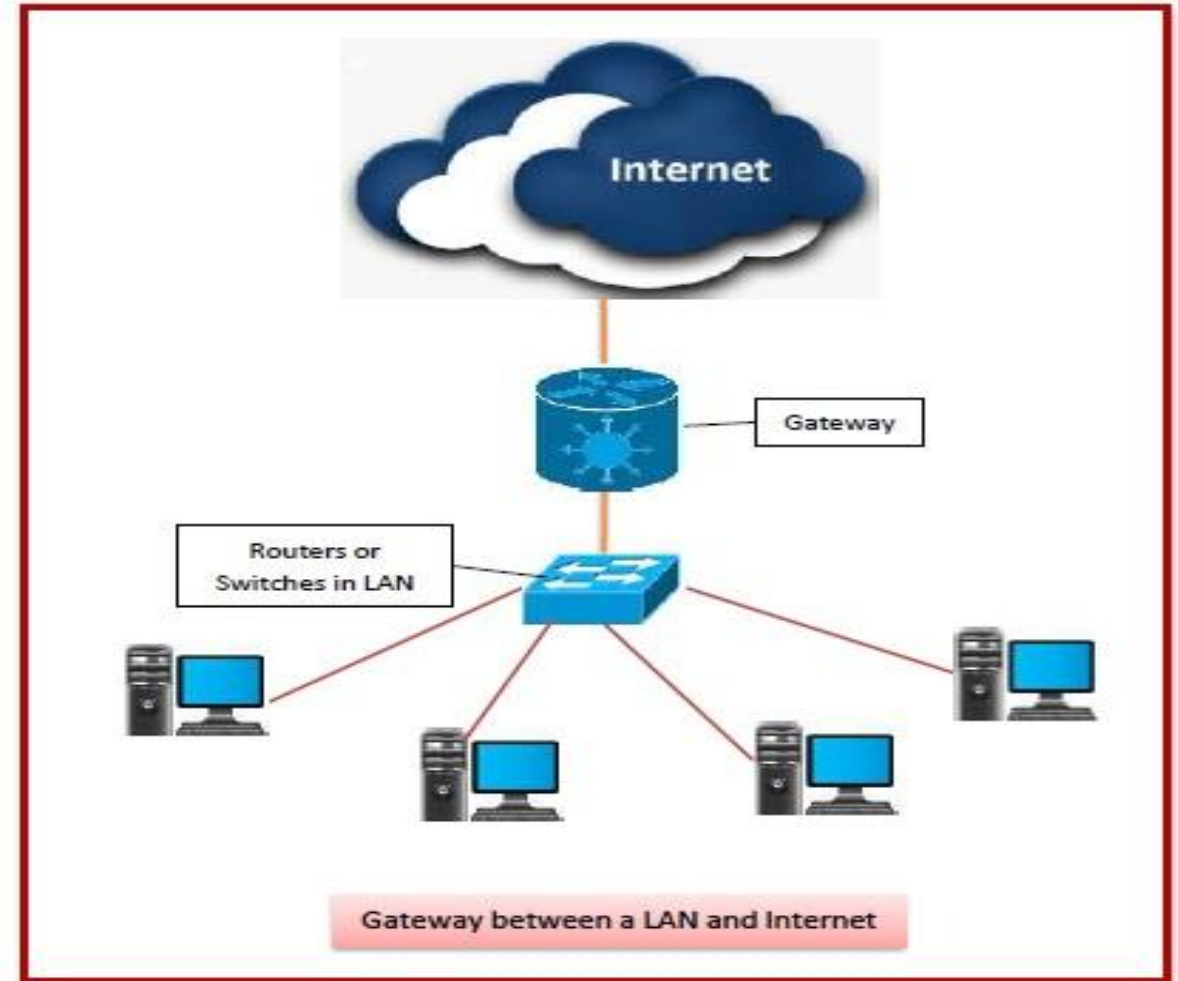
Gateways

- Device that connects dissimilar networks.
- Operates at the highest level of abstraction.
- Expands the functionality of routers by performing data translation and protocol conversion.
- Establishes an intelligent connection between a local network and external networks with completely different structures.
- Gateways serve as an entry and exit point for a network as all data must pass through or communicate with the gateway prior to being routed.
- If a network wants to communicate with devices, nodes or networks outside of that boundary, they require the functionality of a gateway.
- A gateway is often characterized as being the combination of a router and a modem.



Gateways

- A gateway is a network node that forms a passage between two networks operating with different transmission protocols.
- The most common type of gateways, the network gateway operates at layer 3, i.e. network layer of the OSI (open systems interconnection) model.
- However, depending upon the functionality, a gateway can operate at any of the seven layers of OSI model.
- It acts as the entry – exit point for a network since all traffic that flows across the networks should pass through the gateway..



Address Resolution Protocol (ARP)



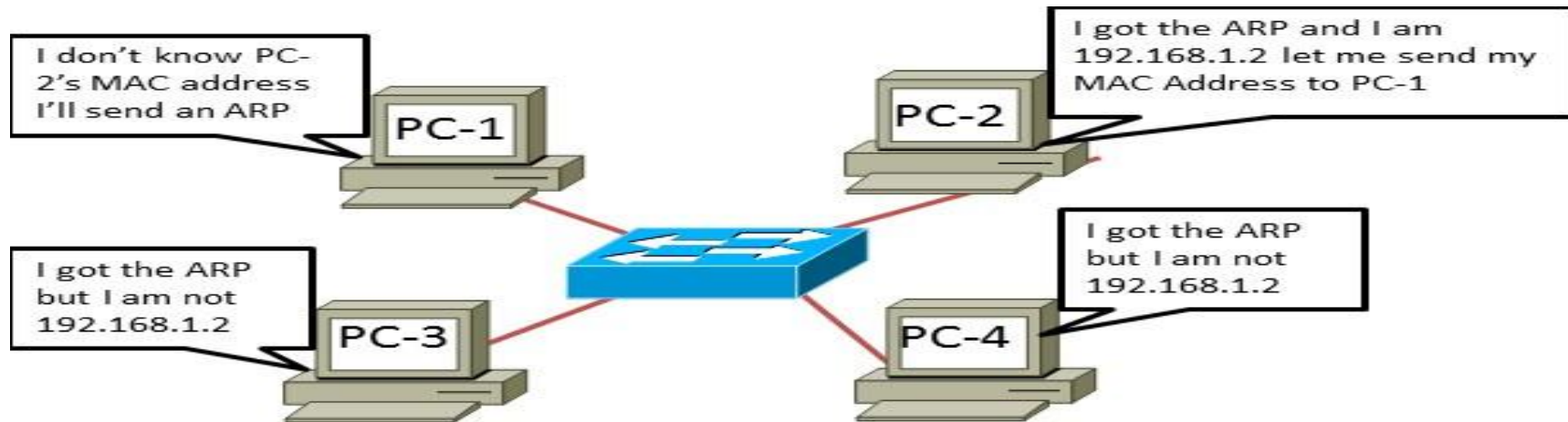
ARP

- Address resolution refers to the process of finding an address of a computer in a network.
- The address is "resolved" using a protocol in which a piece of information is sent by a client process executing on the local computer to a server process executing on a remote computer.
- The address resolution procedure is completed when the client receives a response from the server containing the required address.
- The job of the ARP is essentially to translate 32-bit addresses to 48-bit addresses and vice-versa.
- This protocol works between layer 2 and layer 3 of the OSI model.
- The MAC address resides at layer 2, which is also known as the data link layer and IP address resides at layer 3, this layer is also known as the network layer.



ARP

- Step1 : ARP Broadcast
 - Note: Broadcast is received by everyone and processed by everyone.
 - Step 2: ARP Reply
 - Step 3 : Actual Data Transfer
-
- Router creates an ARP Request message to be sent to all hosts on the subnet.
 - Address resolution protocol message asks "Who has specified IP address ?"
 - Passes ARP request to data link layer process for delivery



Addressing



Addressing



Physical Address/ Link Address

- For example, Ethernet uses a 6-byte (48-bit) physical address that is imprinted on the network interface card (NIC).

Logical Address

- logical address in the Internet is currently a **32-bit address** that can uniquely define a host connected to the Internet. IP address

Port Address

- computer A can communicate with computer C by using TELNET(login into a remote computer). At the same time, computer A communicates with computer B by using the File Transfer Protocol (FTP).

Specific Addresses

- Examples include the e-mail address and any Uniform Resource Locator (URL)



TELNET

- **A network protocol that allows a user on one computer to log into another computer that is part of the same network manage devices over a TCP/IP network.**
- A program that establishes a connection from one computer to another by means of telnet.
- A link established using a telnet program.
- Login into a remote computer using a telnet program.
- **File transfer protocol (FTP)** is an Internet tool provided by TCP/IP.
- It helps to transfer files from one computer to another by providing access to directories or folders on remote computers and allows software, data, text file to be transferred between different kinds of computers.
- **A Uniform Resource Locator (URL)**, is a web address. It is a reference to a web resource that specifies its location on a computer network and a mechanism for retrieving it.



MAC Address / Physical Address/ Ethernet Address

- used on data link layer
- used to identify every NIC uniquely
- is burnt into the ROM part of NIC once written the MAC address can not be changed
- also known as read only address
- to find the MAC address of NIC
 - windows: ipconfig /all
 - linux/macOS: ifconfig
- e.g. 78 : 4f : 43 : 90 : 13 : d0
- size: 6 bytes = $8 \times 6 = 48$ bits
- Group of first three bytes(78 : 4f : 43) represent's manufacturer ID and last 3 bytes (90 : 13 : d0) represents NIC's unique address.
- to find the manufacturer, please visit <https://hwaddress.com/>



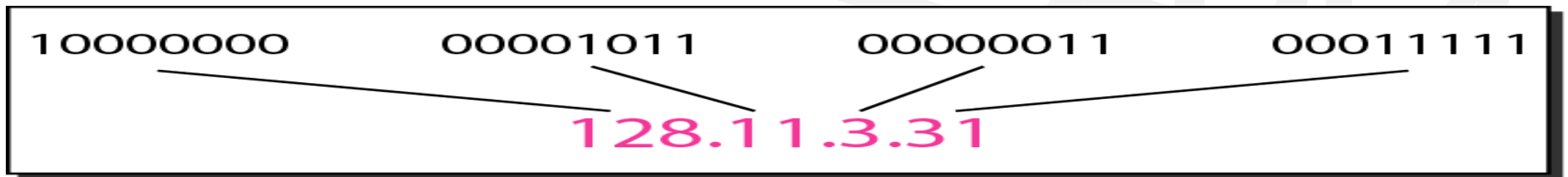
IP Address / Logical Address

- IP address to mean a logical address in the network layer of the TCP/IP protocol suite.
- Identify a machine / device uniquely.
- Size = 4 bytes = 32 bits
- to find the IP address of Machine
 - windows: ipconfig
 - linux/macOS: ifconfig
- IP Versions:
 - IPV4 (32 bits address length)
 - IPV6 (128 bits address length)
- IP addresses are made up of four sets of numbers called **"Octets"**.
- Types
 - Private : used to identify a machine on the LAN and can not be used to connect to internet
 - Public : used to connect to the internet
- e.g.
 - decimal: 192.168.1.6
 - binary : 11000000.10101000.00000001.00000110



IP Addressing Types

- Classful : IP Address is split into 5 classes
- Classless
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than 4 billion)
- **There are two prevalent notations to show an IPv4 address:**
 - binary notation
 - dotted decimal notation



Example

- Find the error, if any, in the following IPv4 addresses.

a. 111.56.045.78

b. 221.34.7.8.20

c. 75.45.301.14

d. 11100010.23.14.67

Solution

- a. There must be no leading zero (045).
- b. There can be no more than four numbers.
- c. Each number needs to be less than or equal to 255.
- d. A mixture of binary notation and dotted-decimal notation is not allowed.



Thank You!!

